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3. Bola

$$\nabla V \text{ bola} = \frac{\partial V}{\partial r}\hat{ar} + \frac{1}{r}\frac{\partial V}{\partial \theta}\hat{a\theta} + \frac{1}{r\sin\theta}\frac{\partial V}{\partial \phi}\hat{a\phi}$$

$V = 3r^2 \sin\theta \cos\theta \sin\phi$ (Volt).

P(observatur) : $r = 2$, $\theta = 60^\circ$, $\phi = 90^\circ$

$\nabla V|_P$ ? $\quad a_{\nabla V}|_P$ : ?

$|\nabla V|_P|$ : ?

Jawab

$$\nabla V \text{bola} = \frac{\partial V}{\partial r}\hat{ar} + \frac{1}{r}\frac{\partial V}{\partial \theta}\hat{a\theta} + \frac{1}{r\sin\theta}\frac{\partial V}{\partial \phi}\hat{a\phi}$$

$$= \frac{\partial(3r^2\sin\theta\cos\theta\sin\phi)}{\partial r}\hat{ar} + \frac{1}{r}\frac{\partial(3r^2\sin\theta\sin\phi)}{\partial\theta}\hat{a\theta} + \frac{1}{r\sin\theta}\frac{\partial(3r^2\sin\theta\cos\theta\sin\phi)}{\partial\phi}\hat{a\phi}$$

$$= \frac{6r\sin\theta\cos\theta\sin\phi}{\partial r}\hat{ar} + \frac{1}{r}\frac{(3r^2\cos\theta\cos\phi)}{\partial\theta}\hat{a\theta} + \frac{1}{r\sin\theta}\frac{(3r^2\sin\theta\cos\theta\cos\phi)}{\partial\phi}\hat{a\phi}$$

$$\nabla V|_P = \frac{6(2)\sin(60^\circ)\cos(60^\circ)\sin(90^\circ)}{\partial r}\hat{ar} + 3(2)\cos(60^\circ) - \sin(60^\circ)\sin(90^\circ)\hat{a\theta} + \frac{1}{\sin 60^\circ} + 3(2)\sin(60^\circ)\cos(60^\circ)\cos(90^\circ)\hat{a\phi}$$

$$= 6\sqrt{3}\cdot\frac{1}{2}\cdot\frac{1}{2}\hat{ar} + 3 - \frac{1}{2}\sqrt{3}\cdot 1\hat{a\theta} + \frac{1}{\frac{1}{2}\sqrt{3}}$$

$3\sqrt{3} \cdot \frac{1}{2} \cdot 0\,\hat{a}_\phi$

$= 3\sqrt{3}\,\hat{a}_r - \frac{3}{2}\sqrt{3}\,\hat{a}_\theta + 0$

$= 3\sqrt{3}\,\hat{a}_r - \frac{3}{2}\sqrt{3}\,\hat{a}_\theta$

$|\nabla V|_{P1} = \sqrt{(3\sqrt{3})^2 + \left(-\frac{3}{2}\sqrt{3}\right)^2}$

$= \sqrt{27 + \frac{27}{4}}$

$= \sqrt{27 + 6.75}$

$= 5.81 \text{ V/m}$

$\hat{a}_{\theta N} = \frac{\nabla V}{|\nabla V|} = \frac{3\sqrt{3}\,\hat{a}_r - \frac{3}{2}\sqrt{3}\,\hat{a}_\theta}{5.81}$

$= 0.89\,\hat{a}_r - 0.45\,\hat{a}_\theta$